How-To Library

Informatica

Configuring Kerberos and SSL for Enterprise Data Catalog

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Abstract

This article provides information about configuring Enterprise Data Catalog in a Hadoop cluster enabled for Kerberos, SSL, or both Kerberos and SSL protocols for authentication and encryption.

Supported Versions

- Enterprise Data Catalog 10.4.1

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Overview

You can deploy Enterprise Data Catalog on an embedded cluster or an existing cluster. The sections include prerequisites and steps that you must complete to deploy and configure Enterprise Data Catalog in a Hadoop cluster enabled for Kerberos, SSL, or both Kerberos and SSL.

An existing cluster is a cluster that you have configured in your enterprise on Cloudera, Hortonworks, or Azure HDInsight.

An embedded cluster is a Hortonworks Hadoop cluster bundled with Enterprise Data Catalog. The Enterprise Data Catalog installer creates an Informatica Cluster Service as an ISP service for an embedded cluster. Enterprise Data Catalog uses Apache Ambari to manage and monitor the embedded Hadoop cluster.

Informatica does not support cross or multi-realm Kerberos authentication. The Enterprise Data Catalog Service principals and the Hadoop service principals must be present in the same realm.

The Informatica domain requires keytab files to authenticate Hadoop services and application services deployed by Enterprise Data Catalog. The keytab files contain the service principal names (SPN) and associated encrypted keys. Create the keytab files before you create cluster nodes and services in the Informatica domain.

For detailed steps and information about installation, see *Informatica Installation for Enterprise Data Catalog* .

Domain Node Hardware Requirements

If you install Enterprise Data Catalog on a Kerberos-enabled existing cluster on Cloudera, Hortonworks, or Azure HDInsight, verify that the domain node satisfies the following minimum hardware requirements:

- Four CPUs with a minimum of four cores.
- 32 GB of memory.
- 60 GB of free hard disk space.

Prerequisites for an Existing Cluster

You can install and deploy Enterprise Data Catalog on an existing Cloudera, Hortonworks, or Azure HDInsight cluster. If you have enabled Kerberos authentication in your enterprise to authenticate users and services on a network, you can configure the Informatica domain to use Kerberos network authentication.

Verify that you complete the following prerequisites before you install and deploy Enterprise Data Catalog on a Kerberos-enabled cluster:

- When you create the Catalog Service for an existing cluster, verify that you configure the following properties:
  - A keytab file that contains all the users in LDAP.
  - Kerberos domain name.
  - HDFS namenode and YARN Resource Manager service principals.
  - Path to Solr keystore file and password.
- Verify that the following prerequisite packages are installed before you enable Kerberos:
  - krb5-workstation
  - krb5-libs
  - krb5-auth-dialog
- Create the following users in the LDAP security domain where <username> is the service cluster name.
  - <username>@KERBEROSDOMAIN.COM
  - <username>/<hostname>@KERBEROSDOMAIN.COM
  - HTTP/<hostname>@KERBEROSDOMAIN.COM
  - Create a keytab file with credentials for all these users created in LDAP. You can create a single keytab file for each user in the KDC server and merge them using the ktutil command to create single keytab file.
  - Create the following folders in HDFS that Enterprise Data Catalog uses as data directories for the Catalog Service: `/Informatica/LDM/<username>` and `/user/<username>`.
  - Change the owner of these two folders to <username>.
  - Create a local user with username as <username> on all the hosts in the cluster. This step is required to launch the application on YARN as the user configured for Catalog Service.
- Set up the `udp_preference_limit` parameter in the `krb5.conf` Kerberos configuration file to 1. This parameter determines the protocol that Kerberos uses when it sends a message to the KDC. Set `udp_preference_limit = 1` to always use TCP. The Informatica domain supports only the TCP protocol. If the `udp_preference_limit` parameter is set to any other value, the Informatica domain might shut down unexpectedly.

Note: Enterprise Data Catalog does not support deployment on a Hortonworks version 2.6 cluster where Kerberos is enabled.

Verify that you complete the following prerequisites before you install and deploy Enterprise Data Catalog on an SSL-enabled cluster:

- Create the directory `<service cluster name>` under `/user` and assign the ownership of the directory to the service cluster user.
- Ensure that you enable SSL for Catalog Service and the domain.
- Import the public certificates of the cluster into the domain truststore.
- Import the public certificates of the Catalog Service and the domain into the cluster truststore.
- For a multiple node deployment, make sure that each solr-keystore file includes the public certificates of solr-keystores on other nodes in the cluster.

- If the Solr keystore and the cluster keystore are different, perform the following steps:
  1. Import all cluster keystores to the Informatica domain truststore.
  2. Import the default keystore from all Informatica domain nodes to the cluster keystore.
  3. Import the cluster keystore to all nodes in the cluster.

Prerequisites for an Embedded Cluster

You can install and deploy Enterprise Data Catalog on an embedded Hortonworks Hadoop cluster.
Verify that you complete the following prerequisites before you install and deploy Enterprise Data Catalog on a Kerberos-enabled cluster:

- Make sure that you merge the user and host keytab files before you enable Kerberos authentication for Informatica Cluster Service.
- Verify that you install the following prerequisite packages before you enable Kerberos:
  - krb5-workstation
  - krb5-libs
  - krb5-auth-dialog

Configure the Informatica Cluster Service

Prerequisites

Verify the following prerequisites before you enable Informatica Cluster Service in a Kerberos-enabled cluster:

- Configure the Key Distribution Center (KDC) hostname and IP address on all cluster nodes and domain machines in the `/etc/hosts` directory.
- Make sure that the krb5.conf file is located in all cluster nodes and domain machines under the `/etc` directory.

Verify the following prerequisite before you enable Informatica Cluster Service in an SSL-enabled cluster:

Ensure that you configure and copy the Informatica domain truststore file to a common location that is accessible to all the cluster nodes.

Configuring Kerberos for Informatica Cluster Service

When you create an Informatica Cluster Service, you can choose to enable Kerberos authentication for the cluster.

Creating an Informatica Cluster Service includes configuring the following properties and options for the service:

1. General properties.
2. Security properties.
3. Hadoop cluster properties.
4. Domain security options.

When you configure the Hadoop cluster properties for the Informatica Cluster Service, you can configure the following options to enable Kerberos authentication for the cluster:

Property	Description
Enable Kerberos Authentication	Select the option to enable Kerberos authentication for the cluster.
KDC Type	Select one of the following Kerberos Key Distribution Center (KDC) types if you had selected the **Enable Kerberos Authentication** option: - Active Directory. Select this option if you want to use Active Directory KDC. - MIT KDC. Select this option if you want to use MIT KDC. Specify the following options after you select the **KDC Type** - KDC Host. Name of the KDC host machine. - Administrator Server Host. The name of the administrator server machine that hosts the KDC server. - Realm. Name of the Kerberos realm on the machine that hosts the KDC server. - Administrator Principal. The Kerberos administrator principal. - Administrator Password. The Kerberos administrator password. - LDAP URL. This property applies to Microsoft Active Directory and represents the URL to the LDAP server directory. - Container DN. This property applies to Microsoft Active Directory and represents the Distinguished Name of the container to which the user belongs.

Troubleshooting Kerberos Issues for an Existing Cluster

The Catalog Service has unexpectedly shut down and the error message in the log file reads as follows: "GSSException: No valid credentials provided (Mechanism level: Server not found in Kerberos database". How do I resolve the issue?

Verify that all the cluster nodes in the `/etc/hosts` domain have fully qualified host names and fix the incorrect host names.

I encounter issues with Kerberos Distribution Center (KDC) availability and see messages similar to the following one: "(java.security.PrivilegedActionException: javax.security.sasl.SaslException: GSS initiate failed [Caused by GSSException: No valid credentials provided (Mechanism level: Connection reset)]) occurred when evaluating Zookeeper Quorum Member's received SASL token. Zookeeper Client will go to AUTH_FAILED state." How do I resolve the issues?

A Kerberos-enabled cluster requires a highly available KDC. Verify that you have enabled high availability for KDC.

Troubleshooting Issues for an Existing Cluster

Consider the following troubleshooting tips when you configure Kerberos for an existing cluster:

The Catalog Service has unexpectedly shut down and the error message in the log file reads as follows: "GSSException: No valid credentials provided (Mechanism level: Server not found in Kerberos database". How do I resolve the issue?

Verify that all the cluster nodes in the `/etc/hosts` domain have fully qualified host names and fix the incorrect host names.

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A Kerberos-enabled cluster requires a highly available KDC. Verify that you have enabled high availability for KDC.

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